

Fig. 7: An actual-size, single-side PCB for the RF-controlled aircraft (transmitter's side)

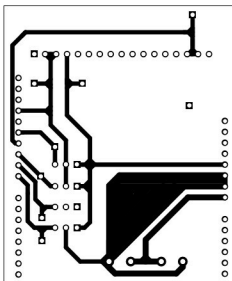


Fig. 9: An actual-size, single-side PCB for the RF-controlled aircraft (receiver's side)

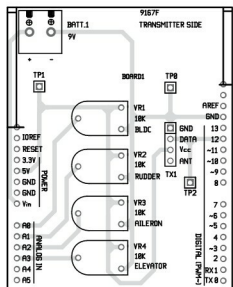


Fig. 8: Component layout for the PCB (transmitter's side)

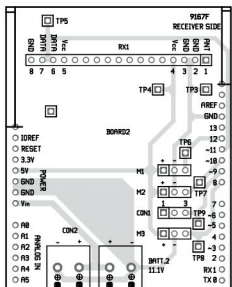


Fig. 10: Component layout for the PCB (receiver's side)

ies) of Arduino IDE before compiling the sketch.

We have used red brick ESC module. If you use any other ESC module, you may be required to modify the delays in the code accordingly.

Construction and testing

An actual-size, single-side PCB for the RF-controlled aircraft transmitter circuit is shown in Fig. 7 and its component layout in Fig. 8. The PCB for the receiver circuit is shown in Fig. 9 and its component layout in Fig. 10.

PCBs of transmitter and receiver are in the form of shields, so mount these PCBs on top of the respective Arduino UNO board. Connect 9V bat-

tery to the transmitter's side and 11.1V battery to the receiver's side. Check the working of BLDC motor by varying VR1, servo motor M1 using VR4, servo motor M2 using VR2 and servo motor M3 using VR3. If there is any problem, verify the test points given in the table. Once this is done, it is time to build the body of the aircraft.

Building a remote-controlled (RC) aircraft requires some basic skills and creativity. There are some rich tutorials available on the Internet.

Some information to build your own RC aircraft can be found from these links:

<http://www.easycrc.com/airplanes/>

EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download at source.efymag.com

<http://rcvehicles.about.com/od/divaircraft/>

<http://www.stenulson.net/rcflight/rcflight.htm>

You can either build the body of the aircraft by yourself or get it from an electronics spare parts shop. The receiver PCB along with the 11.1V battery should be mounted properly on the aircraft. Now, your RF-controlled aircraft is ready to fly.

The aircraft can be launched by gathering speed on a long runway or by launching with the use of hands by literally throwing it into the air. In the first case, as soon as your plane builds up enough speed and catches the wind, it will lift from the ground. But note that if the road surface is uneven, or it is covered with grass, your plane might not be able to gather the necessary speed for takeoff. At this point, you should opt for a hand launch.

Since the aircraft is controlled by a remote control it has a finite range. If your plane gets too far (either vertically or horizontally) from the transmitter in your hand, it might lose power. This will send your plane plummeting back to the ground. The usual way to fly it is in a circular pattern above your head.

After assembling all the parts, switch on both the transmitter and receiver circuits. Slowly vary VR1 to increase the speed of the BLDC motor (propeller). Once it takes off above the ground, vary VR2, VR3 and VR4 and observe the flight pattern. Ensure that it does not go beyond 100m distance from the transmitter in your hand. To bring it back to the ground, reduce the speed by varying VR1. ●

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